

Predicting Accident Severity on UK Road Traffic Data

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Abstract—The issue of road accidents, continues to pose significant public health problems in the UK, where the severity of outcomes is a function of various factors like the type of vehicle, nature of the road, and the surrounding environment. This paper employs a hierarchical data mining methodology on the 2024 UK STATS19 Road Safety dataset, which integrates three relational databases known as Collision, Vehicle, and Casualty into one data analysis database that contains 128,266 records. The dependent variable is road accident severity with three classes: Fatal (2.0%), Serious (25.0%), and Slight (73.0%). Our approach follows a three-step process. To begin with, bias-corrected Cramer’s V is employed in our study for the purpose of feature selection, selecting the best categorical variables that predict severity which include type of vehicle involved, posted speed limit, and junction details. The second step involves evaluating seven models from logistic regression, random forest, and XGBoost using oversampling (SMOTE) to address class imbalance issues and optimizing hyperparameters through iterations. Our random forest model was chosen as the final model, obtaining a test accuracy of 77.48%, a Macro F1 score of 0.6384, and fatal precision of 0.77. Finally, we employ the Apriori algorithm to extract accident risk profiles. The key findings indicate that motorcycles above 500cc and heavy goods vehicles engaging in risky maneuvers on rural roads where the speed is greater than or equal to 60mph are at least two times more probable (Lift = 2.1) to lead to fatalities and severe accidents.

Index Terms—Data Mining, Traffic Accident Severity, Association Rule Mining, Random Forest, UK Road Safety Data.

I. INTRODUCTION

The issue of road safety continues to present a major public health concern within the United Kingdom. With the rising number of vehicles within the cities, the number of variables involved in causing serious accidents becomes more complex. Predicting the severity level of an incident before responding becomes quite essential in ensuring effective response from both the emergency services and transportation authority.

The United Kingdom Department for Transport offers a complete database of such information via its STATS19 Road Safety Open Data set. Despite being useful, such a statistical summary of events fails to capture the full dynamics of road accidents because very few of them are caused by a single factor alone. Instead, severity emerges from dangerous

combinations: a high-displacement motorcycle on an unlit rural road travelling at 60mph, or a heavy goods vehicle executing a risky manoeuvre at an uncontrolled junction. The common univariate analysis approach fails to identify such interactive effects while regression analyses that focus on linear relationships between variables are also not applicable to such complicated scenarios.

This study addresses that gap through a multi-stage data mining framework applied to the 2024 STATS19 dataset, which comprises three relational tables- Collision (100,927 records, 44 variables), Vehicle (183,514 records, 32 variables), and Casualty (128,272 records, 23 variables) merged into a unified analytical dataset of 128,266 records across 96 variables. The target variable is collision severity, classified into three categories: Fatal (2.0%), Serious (25.0%), and Slight (73.0%). This severe class imbalance, identified during initial exploration, is a central technical challenge the study addresses: a model that simply predicts "Slight" for every record would achieve 73% accuracy while being entirely useless for safety applications.

The central research question guiding this work is: How can a multi-stage data mining framework, integrating pattern discovery and predictive modelling, identify and predict the underlying risk factors of road traffic accident severity in the UK?

To answer this, the study pursues two sub-objectives. First, predictive modelling under class imbalance: evaluating whether ensemble learning algorithms can achieve reliable performance on the accident severity classification task, given that the dataset is heavily skewed towards Slight injuries and contains only a small fraction of Fatal cases. Second, interpretable pattern discovery for policy support: determining whether association rule mining, applied to the most predictive features identified by our models, can surface high-risk accident scenarios that are concrete enough to inform policy making and infrastructure improvement decisions.

These two objectives are deliberately sequenced. The process of modelling is where true predictors are determined, thus laying down a statistical foundation for feature selection. The

pattern discovery step comes next and works off this selected feature set, guaranteeing that any risk associations are accurate and not simply a product of randomness. In essence, this is a pipeline from crime data to useful information about safety risks.

In the remainder of this paper, we review and critically evaluate related work on accident severity prediction, association rule mining, and class imbalance handling. This is followed by a description of the dataset, preprocessing steps, and statistical feature selection. We then present our three-phase methodology, modeling results, and comparative analysis across seven model configurations. Subsequently, we report our association rule mining findings and discuss their policy implications. In the end, we outline the study’s limitations and conclude with directions for future work.

II. RELATED WORK

The academic study of road traffic safety has undergone a significant transition from basic descriptive statistics toward advanced data mining frameworks capable of handling the complexity of modern road networks. Taken together, the studies reviewed below shaped three core decisions in our methodology: the choice of ensemble models, the use of scenario-based pattern discovery, and the imperative to handle class imbalance rigorously.

The foundational concept of Association Rule Mining was established by Agrawal et al. [1], who provided the mathematical definitions for support, confidence, and lift. Despite being designed for retail transactions, these metrics can easily be adapted to accident research, given the similar idea behind the use of associations. They played an integral part in Phase Two of our research, which sought to go beyond basic frequencies and reveal combinations of factors related to accidents with dire consequences. Using the lift threshold value of 2.0, we excluded any coincidental association of factors and identified combinations in which the interaction among variables was at least twice as strong as expected by coincidence. In the UK context, association rules have been applied in accident research by Feng et al. [2]. It was shown that the application of such rules could bring into light potential risk associations that regression methods might miss because of assumptions concerning linear relationships. However, the use of old data and exclusion of changes in road vehicle composition and infrastructure makes their findings applicable only in relation to 2024 UK traffic. Nevertheless, one of their results regarding rural high-speed limitations being a key risk factor informed our decision to categorize the data from 2024 according to road type and speed limitations. Our analysis confirmed this pattern, finding that rural high-speed corridors exhibited a lift score of 2.1 for fatal outcomes.

The geographic and infrastructure dimensions of accident severity were explored by Ziakopoulos et al. [3], who compared road crash injuries across different regional networks and emphasised that risk factors are not universal, junction geometry and road infrastructure vary significantly across settings. Although their results are only applicable geographically

to Greece and cannot be generalized to the UK, their key takeaway that context is a driving factor behind severity was part of our rationale for choosing junction-based attributes as some of our main attributes, confirmed by our Cramer’s V test. Comi et al. [4] showed that separating accidents based on type prior to modelling leads to a substantial reduction in output variability, although the clustering process employed by their method requires pre-labeling, thus introducing bias. We, on the other hand, left the segmentation to be done organically by the ARM algorithm, especially motorcycles above 500cc, which have an Fatal/Serious rate of 54.2% in 2024 data.

An overview of existing literature has been done by Santos et al. [5], which through an analysis over 20 years determined that machine learning algorithms, specifically Random Forests, consistently outperform single decision trees and linear regressions when applied to police-reported traffic data. A notable downside according to Santos et al., that is shared in our study, is the use of different countries, severity levels, and time periods, making comparisons between performance unreliable. The superior performance of Random Forests on different types of problems nonetheless makes them a suitable choice for our study. Lin et al. [6] emphasised that rigorous feature ranking is essential to prevent noise from degrading model performance, though their approach relied on deep learning importance scores which sacrifice interpretability. We instead used bias-corrected Cramer’s V, preserving the physical meaning of each feature, critical for producing policy-relevant recommendations. AlMamlook et al. [7] reinforced this direction, finding ensemble architectures substantially outperform Logistic Regression, which struggles with complex variable interactions. Our results replicated this precisely: Logistic Regression achieved a Fatal Precision of just 0.06, while the final Random Forest reached 0.7742.

One of the most significant technical challenges was the extreme sparsity of fatal cases. According to He and Garcia [8], classification methods on imbalance datasets favor the prediction of the majority class and become ineffective for the detection of rare yet important events. In order to solve this problem, SMOTE technique proposed by Chawla et al. [9] was utilized. However, a known limitation of SMOTE is that synthetic interpolation between a small number of real fatal cases can introduce noise rather than genuine signal, a problem we observed directly, as SMOTE improved Fatal Recall by over 13.16% but also contributed to overfitting in more complex configurations, necessitating strict depth constraints. The importance of balanced evaluation is further underscored by a 2025 meta-analysis [10] analysing over two million crash cases, which concluded that Macro F1 is a more appropriate measure than raw accuracy for severity prediction. The final macro F1 score of 0.6384 is consistent with the performance range documented for this type of task within similar research literature, indicative of the complexity involved with predicting road accident severity when faced with such severe class imbalances.

The entire approach was largely informed by Wang et al. [11], who proposed utilizing association rules alongside the

verification process using random forests. Although Wang et al. [11] were concerned specifically with Chinese highway data sets and did not include a discussion on how to deal with the class imbalance issue, the idea of applying a hybrid solution translated well to our case. Specifically, using our Apriori analysis, we found an association between rural areas with 60mph speed limits with a lift of 2.1 while XGBoost placed road speed limits in its top three features independently. The idea of linking logical features was further reinforced by Srivastava et al. [12], which led us to consider only statistically significant features that made intuitive sense.

III. DATASET DESCRIPTION AND EXPLORATION

A. Data Source and Scope

The source code and notebooks for this study are available in this [folder](#) [13]. The study utilises the official STATS19 Road Safety dataset provided by the UK Department for Transport [14], which records all personal injury road accidents reported to the police in Great Britain. The study will be performed for one year (2024) to prevent including obsolete information about the changes in the road situation, features of vehicles, and new constructions appearing from year to year.

B. Dataset Structure and Integration

This dataset consists of three relational tables, combined by the unique indexes Collision Index and Vehicle Reference to produce multidimensional information about each event:

- Collision Table (100,927 records, 44 variables): Contains information about the place and time of an accident (geolocation, speed limits, junction, lighting and weather conditions).
- Vehicle Table (183,514 records, 32 variables): Captures the how like vehicle type, engine capacity, manoeuvre, and driver characteristics.
- Casualty Table (128,272 records, 23 variables): Captures the human outcome, injury severity for every person involved.

After the tables were joined by Collision Index and Vehicle Reference, all the unnecessary administrative columns were deleted and missing values were replaced. The resulting dataset for analysis contained 128,266 observations and 41 variables, among which the target one was collision severity.

C. Statistical Feature Selection

To assess the association between each categorical feature and collision severity, Cramér's V test was employed, with its value lying between 0 and 1. Cramér's V test was also used to find collinear features by calculating their association between each other, especially those connected with junction, road classification, pedestrian, and carriageway categories. Pairs with $V > 0.5$ were considered strongly collinear, and the less informative feature was dropped. Features with low global V but a plausible causal mechanism were further examined through sub-group fatal rate analysis and random forest feature importance before a final retention decision was made.

D. Key Exploratory Findings

There were two important insights drawn from exploratory data analysis which helped form the basis for modeling. The first insight relates to data imbalance where there are more slight injuries compared to serious or fatal ones with slight injuries making up 73.0%, serious 25.0%, and fatal just 2.0%. Second, even at the exploratory stage, a clear high-risk profile emerged: motorcycles over 500cc and HGVs on rural roads with speed limits of 60mph or higher exhibited Fatal/Serious rates exceeding 50%, compared to the 27% dataset baseline, a pattern later confirmed and quantified through association rule mining.

IV. DATA MINING METHODOLOGY

The research adopts CRISP-DM (Cross-Industry Standard Process for Data Mining), where the entire process is structured in six iterative processes: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Given the road safety context, the deployment phase is reinterpreted as the translation of findings into actionable policy recommendations rather than software deployment. The following subsections map our technical work onto this framework.

A. Business and Data Understanding

The business objective was defined as: given a reported road collision, can its severity be predicted reliably, and can the combination of factors that produce severe outcomes be identified in a form useful to transport authorities? This dual objective (predictive and descriptive) motivated the two-phase analytical design. Data understanding began with a thorough exploration of the three STATS19 tables described in Section III, including missing value profiling, distribution analysis, and six exploratory visualisations covering severity by speed limit, light conditions, urban/rural classification, hour of day, and numeric feature correlations. These charts confirmed the class imbalance problem and surfaced the first indication that rural high-speed roads and specific vehicle types carry disproportionate risk.

B. Data Preparation

Data preparation involved several sequential steps. Columns with more than 50% missing values were dropped, including historic duplicate fields retained in the STATS19 schema for legacy compatibility. Sentinel values encoded as -1 were replaced with NaN and subsequently imputed: special conditions at site was filled with code 0 (None) per the data dictionary, second road class with code 6 (Unclassified), age fields were imputed using group-wise median by severity class, and remaining categorical nulls were filled with an Unknown category. Values of 9 and 99, which denote unknown responses in the STATS19 coding scheme, were also recoded as Unknown to prevent them from being treated as numeric signals.

Coordinate duplicates (easting/northing and latitude/longitude) were confirmed as redundant via Pearson

correlation above 0.999 and one pair was dropped. Age band columns were dropped in favour of raw age given a Pearson correlation of 0.98 with their numeric equivalents. Near-zero variance columns including towing and articulation (96.7% single category), vehicle left hand drive, bus or coach passenger (98.4% single category), and pedestrian road maintenance worker (100% single category) were removed. Vehicle direction columns were dropped following a Cramer's V of 0.098 with severity and high mutual collinearity of 0.817. Trunk road flag was removed due to a Cramer's V of only 0.078 with severity and a 91.8% concentration in a single category.

Date and time features were engineered into cyclical representations: hour of day was decomposed into sine and cosine components to preserve continuity across midnight, and month and weekend flag were extracted from the date field. Label Encoder was used to transform categorical variables, and this method fits tree-based algorithms since it assumes no ordinality between them. The data set had 128,266 records with 41 features. We performed a stratified train-test split of 80/20, resulting in 102,612 training and 25,654 testing instances.

C. Statistical Feature Selection through Cramer's V (Phase 1)

Before modelling, a combination of Pearson correlation and Cramér's V was used to identify redundant and weakly predictive features. For numeric variables, Pearson correlation was computed to detect redundancy, the OSGR easting and northing coordinates were found to be near-perfectly correlated with longitude and latitude respectively ($r > 0.99$) and were therefore removed. Similarly, age band features were confirmed to be direct binned encodings of their corresponding raw age columns ($r > 0.90$) and were dropped in favour of the continuous representations. For categorical predictors, Cramér's V was used to measure both the association with collision severity and pairwise collinearity between conceptually related features. Cramér's V produces a symmetric score between 0 and 1, where 0 indicates no association and 1 indicates perfect association. Feature pairs with $V > 0.5$ were considered to exhibit meaningful collinearity, with the less informative feature removed. Features with low global V but a strong causal rationale, specifically the carriageway impact features, were further examined through sub-group fatal rate analysis and random forest feature importance before a final retention decision was made.

D. Predictive Modelling Under Class Imbalance (Phase 2)

The modelling phase evaluated seven configurations across three algorithm families to identify the best balance between overall accuracy and minority class sensitivity. Logistic Regression was included as a linear baseline. Random Forest and XGBoost were selected as the primary ensemble candidates based on their established performance in traffic safety literature. All baseline models used class weight balancing as a first-line imbalance correction.

Baseline evaluation revealed two problems. The baseline Random Forest model delivered a Train Macro F1 score of 0.96 against a Test Macro F1 score of 0.64, demonstrating a high level of overfitting due to an unlimited tree depth and high dimensionality. In all three baseline models, Fatal Precision proved to be extremely low, thus verifying the necessity of class weighting due to the minority class being less than 2%.

Three improvement strategies were introduced in response. Feature selection reduced the input space to the top 20 features ranked by XGBoost importance, chosen over Random Forest importance because XGBoost exhibited the smaller train-test gap, making its importance estimates more reliable. Stricter hyperparameter constraints were applied to Random Forest, including reduced maximum depth and increased minimum samples per leaf, to limit memorisation of training patterns. SMOTE oversampling was applied to the training set only, never to the test set, to generate synthetic Fatal and Serious instances and provide the model with a more balanced learning signal. PCA was deliberately excluded as a dimensionality reduction option because it would destroy the physical interpretability of features, which is essential for producing policy-relevant outputs.

RandomizedSearchCV with three-fold cross-validation was used for hyperparameter tuning across all model variants, optimising for Macro-F1 rather than accuracy to avoid rewarding majority-class prediction. The final model selection prioritised Fatal Precision alongside Macro-F1, as false positive fatal predictions waste emergency resources and undermine operational trust in the system.

E. Interpretable Pattern Discovery via Association Rule Mining (Phase 3)

Following model evaluation, association rule mining was conducted to surface interpretable high-risk scenarios for policy use. The top 25 features ranked by XGBoost importance were manually curated to 16 interpretable variables, removing casualty type (too granular), hour cosine (opaque to non-technical audiences), and sex of casualty (redundant with sex of driver in most records). Each retained feature was binarised into a high-risk versus other indicator using the global Fatal/Serious rate of $\approx 27\%$ as the threshold, so that a feature value was flagged as high-risk only if its associated Fatal/Serious rate exceeded the dataset baseline.

The Apriori algorithm was run with a minimum support of 0.01, chosen to capture rare but meaningful fatal combinations without generating trivial rules. Rules were filtered to include only Fatal or Serious severity as the consequent, ensuring all discovered patterns are directly policy-relevant. The top 10 rules were ranked by lift, and antecedent frequencies were counted across these rules to identify the most consistently appearing risk factors. It was also necessary to break down the vehicle category to see what sub-categories were causing the risky driving indication.

**16 Selected Features for Association Rule Mining
(curated from XGBoost top-25 importance ranking)**

Vehicle & Driver (7 features)		Road & Environment (5 features)		Collision & Casualty (4 features)	
vehicle_type	#17				
vehicle_manoeuvre	#6				
vehicle_leaving_carriageway	#5	urban_or_rural_area	#1		
skidding_and_oversturning	#16	speed_limit	#2	first_point_of_impact	#3
number_of_vehicles	#9	first_road_class	#11	hit_object_off_carriageway	#8
sex_of_driver	#4	junction_location	#12	casualty_class	#10
age_of_driver	#13	light_conditions	#23	age_of_casualty	#15

Fig. 1. The 16 features selected for association rule mining, curated from the XGBoost top-25 importance ranking. Numbers indicate XGBoost importance rank. Features are grouped into three semantic categories: Vehicle & Driver, Road & Environment, and Collision & Casualty.

F. Ethical Considerations

STATS19 is a completely anonymous data set that is made freely available under the terms of the Open Government Licence by the UK Department for Transport. It would not be possible to identify an individual through any of the variables employed in this study since the data set contains no names or addresses. The acknowledgement of AI assistance for grammatical enhancement is disclosed transparently in the paper, with all analytical contributions being the original work of the research team.

V. RESULTS

This section presents the findings from all three phases of the framework, evaluates the modelling strategy, and discusses the implications of the results for road safety policy. The evaluation approach was designed to go beyond raw accuracy, recognising that a classifier predicting Slight for every record would achieve 73% accuracy while providing no actionable value.

A. Evaluation Strategy

Considering the severe skewness of the dataset, accuracy is insufficient as a means of measuring model performance. The main metric adopted for this task is Macro F1-score. As per the name, Macro F1 score calculates F1 score individually for each category and averages them, not taking into account the class weight, ensuring the same importance of the performance on Fatal category, which makes up only 2% of records. Fatal Precision was additionally tracked as a secondary metric, as false positive fatal predictions have direct operational consequences: incorrectly flagging a collision as fatal misallocates emergency resources. Fatal Recall was monitored but treated as a secondary concern, since the cost of a missed fatal case, while serious, is partially mitigated by the fact that all

serious collisions receive emergency response regardless. This parameterisation of the evaluation strategy reflects a deliberate policy-facing design choice rather than a purely technical one.

B. Baseline Model Performance

The three baseline models namely Logistic Regression, Random Forest, and XGBoost, were evaluated using class weight balancing as the sole imbalance correction. Logistic Regression achieved a Test Accuracy of 0.51 and a Fatal Precision of just 0.06, confirming that a linear model cannot capture the complex, non-linear interactions between variables such as vehicle type, speed limit, and road class. Base Random Forest showed higher performance, obtaining 0.64 Test Macro-F1 and 0.7742 Fatal Precision, but demonstrated a very high level of overfitting. For example, its Train Macro-F1 score was 0.96, thus creating a gap of 0.32, which is due to the absence of tree depth limitations and presence of large dimensions. The base XGBoost achieved a Test Macro-F1 of 0.50 with a smaller train-test gap, suggesting better generalisation but weaker overall discrimination. These results confirmed that class weight balancing alone was insufficient and that further intervention was required.

C. Model Improvement and Parameterisation

Three improvement strategies were applied iteratively. Feature selection using the top 20 XGBoost importance scores reduced the input space and partially addressed overfitting in RF v2, but reduced the Test Macro-F1 to 0.52, indicating that the removed features, despite their lower individual importance, contributed useful collective signal. Adding SMOTE to RF v2 improved Fatal Recall substantially but did not recover the Macro-F1 loss, and introduced instability in the form of inflated training metrics. As expected, SMOTE worked best when applied together with XGBoost: version three of XGBoost, using the entire feature set after applying SMOTE

TABLE I
COMPARATIVE ANALYSIS OF METHODOLOGIES AND STRATEGIC ADVANTAGES

Method	Key Finding / Result	Strategic Advantage ("Why it is better")
Cramer's V	Identified high-impact features including Vehicle Type, Speed Limit, and Junction Detail.	Appropriate for nominal and ordinal categorical data regardless of scale; provides a symmetric, bounded effect size measure (0-1) that is interpretable without assuming any distributional form, unlike correlation-based measures which require numeric inputs.
Apriori	Discovered "High-Risk Profiles" with a Lift score of 2.1.	Better for Discovery: Unlike standard statistics, it uncovers how factors <i>interact</i> (e.g., Rural + 60mph) to create hazardous scenarios.
Random Forest	Achieved an overall predictive accuracy of 77.5% .	Better for Robustness: The ensemble nature handles non-linear relationships and is less prone to overfitting than a single decision tree.
SMOTE	Substantially improved XGBoost's Fatal Precision (0.14 → 0.62), but degraded Random Forest's Macro-F1 (0.64 → 0.51).	Better for revealing model-specific behaviour: Comparative testing exposed that synthetic oversampling helps boosting models but harms bagging models, informing the final model-specific imbalance strategy.

technique, scored the highest Test Accuracy value of 0.7845, obtaining 0.5936 Macro-F1 score; however, its Fatal Precision value of 0.6245 is lower than Random Forest's.

The choice of SMOTE parameters warrants discussion. A standard 1:1 oversampling ratio was applied, generating synthetic Fatal instances until class proportions were equalised in the training set. A more conservative ratio would have reduced the risk of the model learning synthetic noise, while a more aggressive ratio might have further improved recall at the cost of precision. In practice, the strict depth constraints applied alongside SMOTE were necessary to prevent the model from fitting to interpolated artefacts rather than genuine fatal patterns.

D. Final Model Selection

The Base Random Forest model was chosen as the final model due to its better Fatality Precision and Macro-F1 score of 0.7742 and 0.6384 respectively. Even though XGBoost v3 performed better accuracy wise, the operational scenario of using a prediction algorithm in a real-world setting where false fatality predictions could lead to unnecessary precautions indicates that the base random forest model should be used due to its lower number of false fatalities. The full comparative results across all seven configurations are presented in Table II.

E. Dominant Risk Factors

The XGBoost feature importance ranking was used to identify the most influential predictors of severity, and these features in turn formed the candidate pool for our subsequent association rule mining. Vehicle type emerged as the single most influential predictor of severity. Breaking down the vehicle types with high-risk flags showed that motorcycles above 500cc had a rate of 54.2% for fatality/seriousness from 4,536 collisions, almost twice the 27% benchmark rate of the entire dataset. Electric motorbikes were at 47.5%, motorbikes ranging between 125-500cc were at 40.6%, and HGVs over 7.5 tonnes were at 44.2%. This gradient across vehicle categories suggests that vehicle mass and the level of physical protection

afforded to the road user are key contributors to outcome severity. Combined with the behavioural and environmental factors identified in the subsequent association rule mining, these findings provide a direct statistical basis for tiered licensing and mandatory safety training policies for higher-displacement motorcycles, as well as stricter operational regulations for heavy goods vehicles.

A similar disaggregation of the high-risk vehicle manoeuvre flag, which appeared as an antecedent in 7 of the top 10 ARM rules, pinpointed overtaking a moving vehicle on its offside as the most dangerous specific action, with a Fatal/Serious rate of 43.9% (n=2,270), well above the 27% baseline. Other lane-changing and overtaking manoeuvres followed at 30-31%. The clustering of risk around overtaking and lane-changing actions is consistent with the broader rural high-speed archetype, since these manoeuvres demand sight distance and lane discipline that single-carriageway rural roads often lack, providing a more concrete behavioural target for road safety interventions than the aggregated manoeuvre flag alone.

F. Association Rule Mining Results

The Apriori algorithm, run with a minimum support of 0.01 and filtered to Fatal/Serious consequents, produced a set of rules whose top 10 by lift converged on a single dominant risk archetype. Vehicle type high-risk appeared as an antecedent in all 10 rules, confirming its status as the strongest individual risk factor. Rural area appeared in 8 of the 10 rules, high-risk first point of impact in 8, high-risk vehicle manoeuvre in 7, high-risk road class in 6, and high speed limit in 6. When these factors co-occur, the Fatal/Serious probability reaches 55-58%, representing a lift of approximately 2.1 over the 27% baseline.

The minimum support threshold of 0.01 was chosen to balance coverage and specificity for the 2024 dataset: a sufficiently low value to retain rare but policy-relevant Fatal/Serious combinations, while still ensuring each surviving rule covered enough collisions to be actionable.

Read alongside the EDA, the ARM results also help interpret the 70mph anomaly observed earlier: severity at 70mph

TABLE II
COMPARATIVE PERFORMANCE OF PREDICTIVE MODEL ITERATIONS

Model Variation	Accuracy	Macro-F1	Fatal Recall	Fatal Precision	Impact / Role
Logistic Regression	0.5122	0.3712	0.6853	0.0584	Baseline; high false positive rate for Fatal class.
Random Forest (Base)	0.7748	0.6384	0.3825	0.7742	Selected model; best Fatal Precision and Macro-F1 balance.
XGBoost (Base)	0.6513	0.4964	0.6753	0.1436	High sensitivity but lower precision.
RF v2 (FS + Depth)	0.6701	0.5211	0.5120	0.2180	Optimized for generalizability.
RF v2 + SMOTE	0.7013	0.5077	0.4303	0.1885	Enhanced minority detection.
XGBoost v2 (FS+SMOTE)	0.7609	0.5285	0.3048	0.3469	Improved overall stability.
XGBoost v3 (SMOTE)	0.7845	0.5936	0.3446	0.6245	Highest raw predictive power.

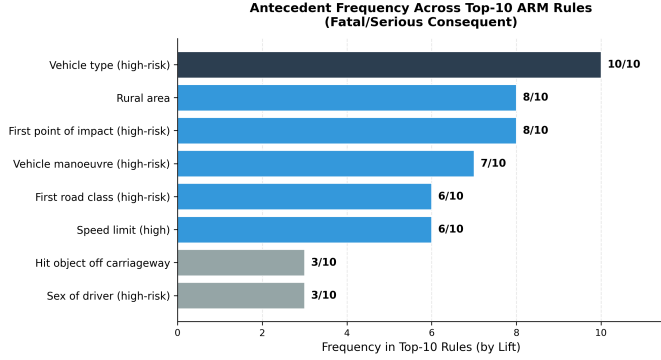


Fig. 2. Antecedent frequency across the top-10 association rules ranked by lift (Fatal/Serious consequent). Dark bars indicate factors appearing in all 10 rules; blue bars indicate factors appearing in 6–8 rules; grey bars indicate factors appearing in fewer than half the top-10 rules.

was marginally lower than at 60mph despite the higher speed. The ARM rules show that high-risk severity outcomes cluster on rural, high-speed roads rather than on motorways, which is consistent with the hypothesis that motorway-grade infrastructure barriers, lane separation, and controlled access provide measurable protective benefits. Together, these two pieces of evidence suggest that the danger on 60mph rural roads is not speed alone but the absence of protective infrastructure at those speeds, pointing toward targeted infrastructure investment rather than a blanket speed reduction policy.

G. Implications of Findings

Taken together, the modelling and ARM results answer the two research sub-objectives directly.

On predictive modelling under class imbalance: our experiments demonstrated that reliable Fatal-class performance is achievable on severely imbalanced police-reported data, but only when imbalance handling is matched carefully to the model family. Class weighting proved most effective for Random Forest, achieving the best Fatal Precision (0.77) without sacrificing overall stability. SMOTE, in contrast, was harmful when applied to Random Forest but transformed XGBoost from a low-precision, high-recall classifier into a high-precision one, Fatal Precision rose from 0.14 to 0.62, at the cost of reduced recall, repositioning the model as a more conservative but trustworthy detector. We also found that model selection must be guided by minority-class-sensitive

metrics such as Fatal Precision and Macro-F1 rather than overall accuracy, since the 73% Slight majority makes accuracy a misleading indicator on this dataset.

On interpretable pattern discovery: the ARM pipeline, applied to features curated from the XGBoost importance ranking, successfully surfaced a concrete, policy-relevant risk archetype, high-displacement motorcycles and heavy goods vehicles performing risky manoeuvres on rural roads with speed limits of 60mph or higher, that is specific enough to guide targeted infrastructure investment, licensing reform for high-powered motorcycles, and stricter operational regulations for heavy goods vehicles.

VI. CONCLUSION

This project demonstrated that traffic accident severity is not a random occurrence but follows distinct, predictable patterns. By integrating statistical testing, association rule mining, and ensemble learning, we moved road safety analytics from descriptive crash statistics toward proactive, evidence-based risk management. The framework identified a stable Fatal/Serious risk archetype with a lift of approximately 2.1, high-displacement motorcycles and heavy goods vehicles operating on rural high-speed corridors, giving traffic authorities a concrete, statistically grounded target for infrastructure investment and policy reform. The Random Forest model's high Fatal Precision (0.77) further shows that machine learning can flag vehicle-specific risks reliably enough to support resource allocation decisions without overwhelming analysts with false positives. Together, these findings provide a blueprint for transforming raw police records into actionable safety intelligence. Future work should focus on retraining the models exclusively on pre-event features and integrating real-time signals such as live weather, traffic density, and driver behaviour data, in order to move the framework from post-hoc analysis toward genuine collision prevention.

A. Limitations

While our multi-stage framework yields useful predictive performance and interpretable risk patterns, several limitations should be acknowledged.

- Data scope: The study relies exclusively on police-reported STATS19 data from a single year (2024). There is a risk that minor accidents may be underreported in police reports, leading to possible selection bias,

while the one-year time period makes it impossible to examine safety trends beyond a single year. Moreover, the lack of detailed human factors, such as distraction, fatigue, alcohol consumption, and vehicle condition, as well as contextual features like live traffic volume and microclimate, can be expected to significantly impact collision severity, even though they are not included in the available police reports.

- Class imbalance handling: Although considerable effort was put into exploring different class weighting techniques and SMOTE, the minority Fatal class continues to pose some challenges for the model. Specifically, while the SMOTE algorithm proved to be detrimental to the Random Forest classifier's Macro-F1 score (it dropped from 0.64 to 0.51 due to the fact that artificial data seemed to influence tree split criteria), it was highly beneficial to the XGBoost Fatal Precision score (it increased from 0.14 to 0.62).
- Post-event features: Several of the most predictive features in our final models: `first_point_of_impact`, `hit_object_off_carriageway`, `vehicle_leaving_carriageway`, and `skidding_and_overturning`, are post-event variables that can only be observed after a collision has occurred. The rules are applicable in retrospective analysis, as well as for policies aimed at mitigating the risk, but not for actual risk prediction in real time due to the lack of retraining of the models with purely pre-casual features.
- Correlation, not causation: The association rules found are purely correlative in nature; their presence in an accident does not imply causation. The presence of rural roads, high speed limits, and specific vehicles may indicate that these factors correlate in accidents of higher severity but cannot prove causation, as would be needed for causal inference in experimental settings.

B. Future Work

Several extensions would meaningfully improve upon this study. Retraining the models exclusively on pre-event features like speed limit, road class, light conditions, vehicle type, and time of day would establish whether reliable severity prediction is feasible before a collision occurs, which is the threshold for genuine real-time deployment. Enriching the dataset with real-time signals such as live weather feeds, traffic flow data, and telematics would address the human factors gap and likely improve Fatal Recall without sacrificing precision. Considering more than one year would enable the framework to be able to observe trends, which in turn will make it possible to determine whether any of the risk archetypes that have been identified are stable over time or not. Lastly, by utilizing causal inference techniques like propensity score matching or instrumental variable regression analysis, it would be possible to test the causality of the relationship between the factors observed in the ARM framework.

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